ARENA OS



Smart Stadium
Real-Time Intelligence

“See. Predict. Respond.”

Executive Summary

The problems

- 20% revenue lost due to long concession lines (fans give up)
- 45+ minutes lost in parking & exit traffic per event.
- High stress levels reported by families due to overcrowding.

Our proposed solution

- queue-aware food alerts.
- Real-time routing for parking and restrooms.
- enhanced user experience

The value

- Increase sales, reduce costs
- Prevent congestion before it happens
- Deliver a premium fan experience

The Explosive Growth of the Sports Tourism Sector

Growth in Saudi-Sports Tourism - 5 Year Intevals

Projected Figures

Year 2005	Year 2010	Year 2015	Year 2020	Year 2025	Year 2030	Year 2035
No. of sports tourists~30,000	No. of sports tourists~50,000	No. of sports tourists~80,000	No. of sports tourists~50,000	No. of sports tourists~1.2 mil	No. of sports tourists~3.5 mil	No. of sports tourists~6.0 mil
Tourist spend ~0.3 bn	Tourist spend ~0.6 bn	Tourist spend ~1.0 bn	Tourist spend ~0.6 bn	Tourist spend ~15 bn	Tourist spend ~45 bn	Tourist spend ~80 bn
GDP Impact ~0.4 bn	GDP Impact ~0.9 bn	GDP Impact ~1.4 bn	GDP Impact ~0.8 bn	GDP Impact ~22 bn	GDP Impact ~70 bn	GDP Impact ~120 bn

Estimates based on Ministry of Sport disclosures, major event economic impact studies, and tourism multiplier benchmarks.

The Current Fan Journey: A Cycle of Frustration

- **Scenario:**

Ahmed arrives at a sold-out stadium excited for the game.

He spends 25 minutes circling the venue looking for parking.

Misses the kickoff while stuck in long security and bathroom lines.

Leaves his seat early to grab food, only to face another 20-minute queue.

After the event, everyone exits at once roads are gridlocked, frustration is high.

- **The Result:**

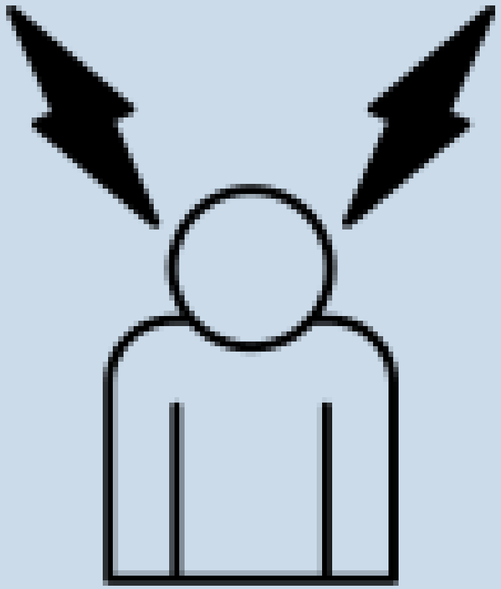
Fans feel stressed instead of entertained

Vendors lose sales due to long queues

Venues face congestion, safety risks, and poor satisfaction scores

- **The Core Problem:**

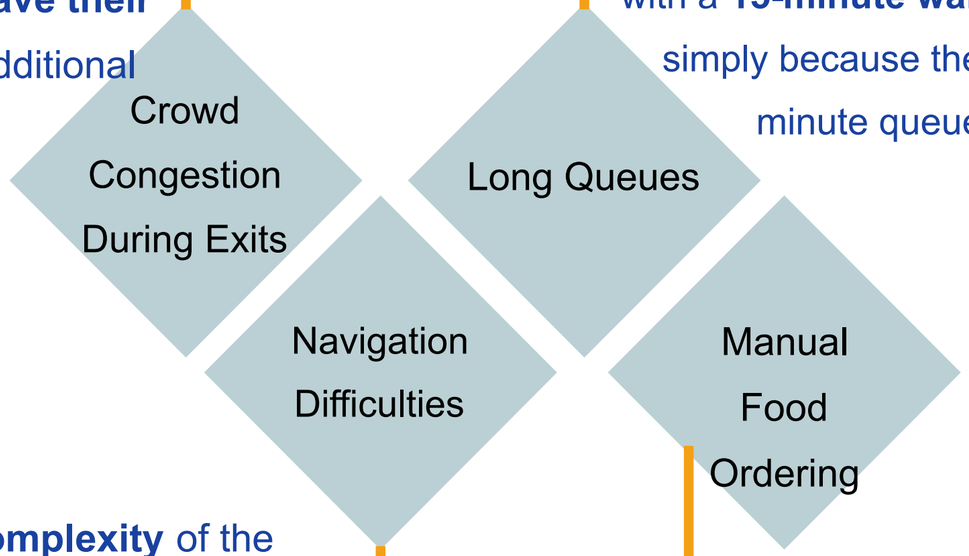
Crowds move blindly with no real-time intelligence, no prediction, and no coordination.



What Problems Do Stadium Visitors Face

If a fan's first attempt to leave their seat results in getting lost, they are **60% less likely to leave their seat** again to purchase merchandise or additional food.

Research shows that **75% of fans** will choose a queue with a **15-minute wait** if it is the closest one to their exit, simply because they lack the data to know that a 2-minute queue exists **45 seconds away**.



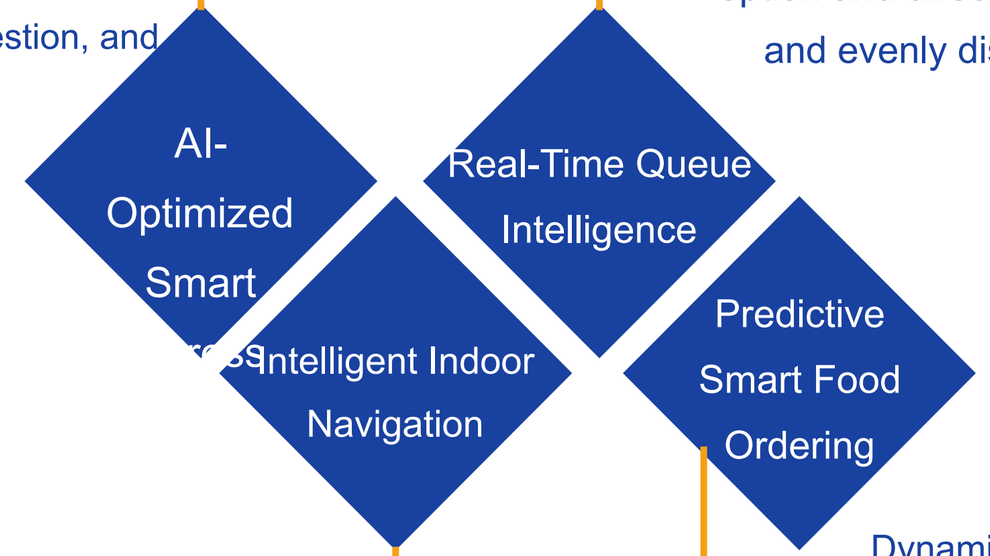
The large scale and **architectural complexity** of the venue can turn a simple trip into a **stressful** navigation experience.

Revenue Loss: For a **50,000-seat stadium**, a **10% reduction** in queue wait times can lead to an estimated **15% to 20%** increase in total
food and beverage revenue

How Does Arena OS Solve These Problems?

AI dynamically monitors crowd density and movement patterns across the stadium, then adjusts exit gates in real time to prevent bottlenecks, reduce congestion, and ensure a safer, smoother flow of people.

AI analyzes live queue data to recommend the fastest option and direct fans accordingly, reducing wait times and evenly distributing crowds across the venue.



Smart signage and in-app directions that update in real time based on live crowd conditions, helping fans navigate the stadium efficiently while avoiding congestion and delays.

Dynamic menu recommendations tailored to a fan's location and crowd conditions, combined with mobile pre-ordering and timed pickup for a faster, smoother experience.

Growth and fan loyalty

reducing queue times by **25 percent** using **dynamic queue** suggestion

creating **unique experience** through memorable collectibles



Implementing Real-Time Point-of-Sale **(POS)Personalization** to drive a **30% uplift** in per-capita ancillary **revenue**.

increasing customer retention rate by **70 percent** thorugh experience satisfaction

relieving crowd congestion by **70 percent** using thorugh **dynamic routing**



The Journey to Implementation

Implementaation Strategy

Fan Retention

Smart Stadium Environment

Distributed IoT Infrastructure

Next-Gen Fan Experience

phase 1 → phases phase 2 → phase 3

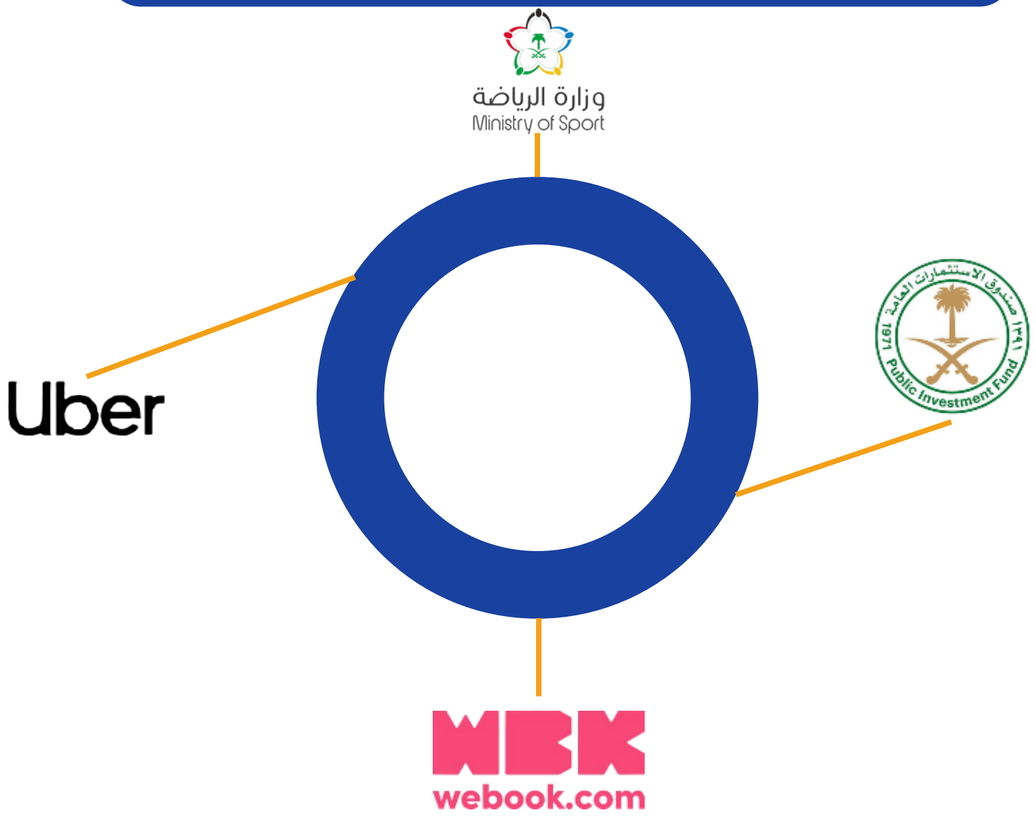
Arena OS Technology Stack

- Block Chain
- NFT
- IOT
- AI/ML

- Block chain enhanced nft tokens as unique and permanent collectibles
- Machine learning for dynamic data analysis
- Integrated iot sensors to fetch live data
- Artificial intellegince for dynamic suggestion tracking and customer satisfaction

The Partners in our Success

Strategic Collaborators



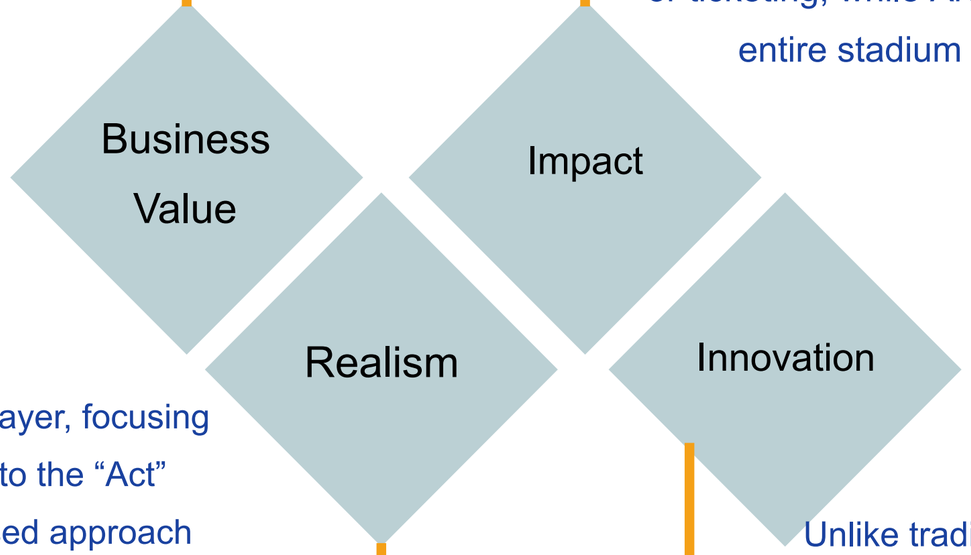
Roles & Contributions

- Funds smart stadium intergration
- Aligns with Vision 2030 goals
- Integrates ticketing, smart parking, and gate-based access
- Implements a reward system (booking points, loyalty benefits, event perks)
- Manages smart drop-off zones
- Reduces traffic congestion
- Oversees federations, Saudi Olympic & Paralympic Committee, and Sports for All
- Enables nationwide adoption and ensures safety regulation

The Path that Marks our Success

The platform isn't limited to sports. The "Living System" model can be licensed to airports, shopping malls, and theme parks, creating a Software as a Service (SaaS) model with high recurring profit margins.

Most solutions address a single function, such as parking or ticketing, while Arena OS connects and optimizes the entire stadium as one integrated ecosystem.



Implementation can begin with the “Sense” layer, focusing on data collection, and gradually expand to the “Act” features, such as AR navigation. This phased approach allows the system to learn and adapt to the unique crowd patterns and behavior of each stadium.

Unlike traditional stadium systems, Arena OS combines real-time crowd intelligence, predictive AI, and adaptive guidance into one unified, self-optimizing platform that anticipates congestion and dynamically adjusts navigation, exits, and services to enhance safety and efficiency.

Business value

Scalability & Revenue Model

- Not limited to sports venues; adaptable to airports, malls, and theme parks
- “Living System” model licensed as a high-margin SaaS platform
- Enables smart parking, personalized navigation, and commercial optimization
- Unlocks digital extensions such as collectibles and premium experiences
- Creates diversified revenue streams through vendor, ticketing,



Innovation

Innovation – Arena OS

- AI-powered stadium orchestration platform
- Uses real-time crowd sensing and predictive analytics
- Anticipates congestion and operational risks before they escalate
- Dynamically optimizes movement, exits, queues, and service distribution
- Integrates fans, operators, and vendors into one connected ecosystem
- Delivers safer operations, higher efficiency, and a seamless fan experience



Impact

Integrated Ecosystem Advantage

- Moves beyond single-function solutions like parking or ticketing
- Connects all stadium operations into one unified platform
- Optimizes movement, services, and crowd flow holistically
- Enables real-time coordination across the entire venue
- Delivers a seamless, end-to-end stadium experience



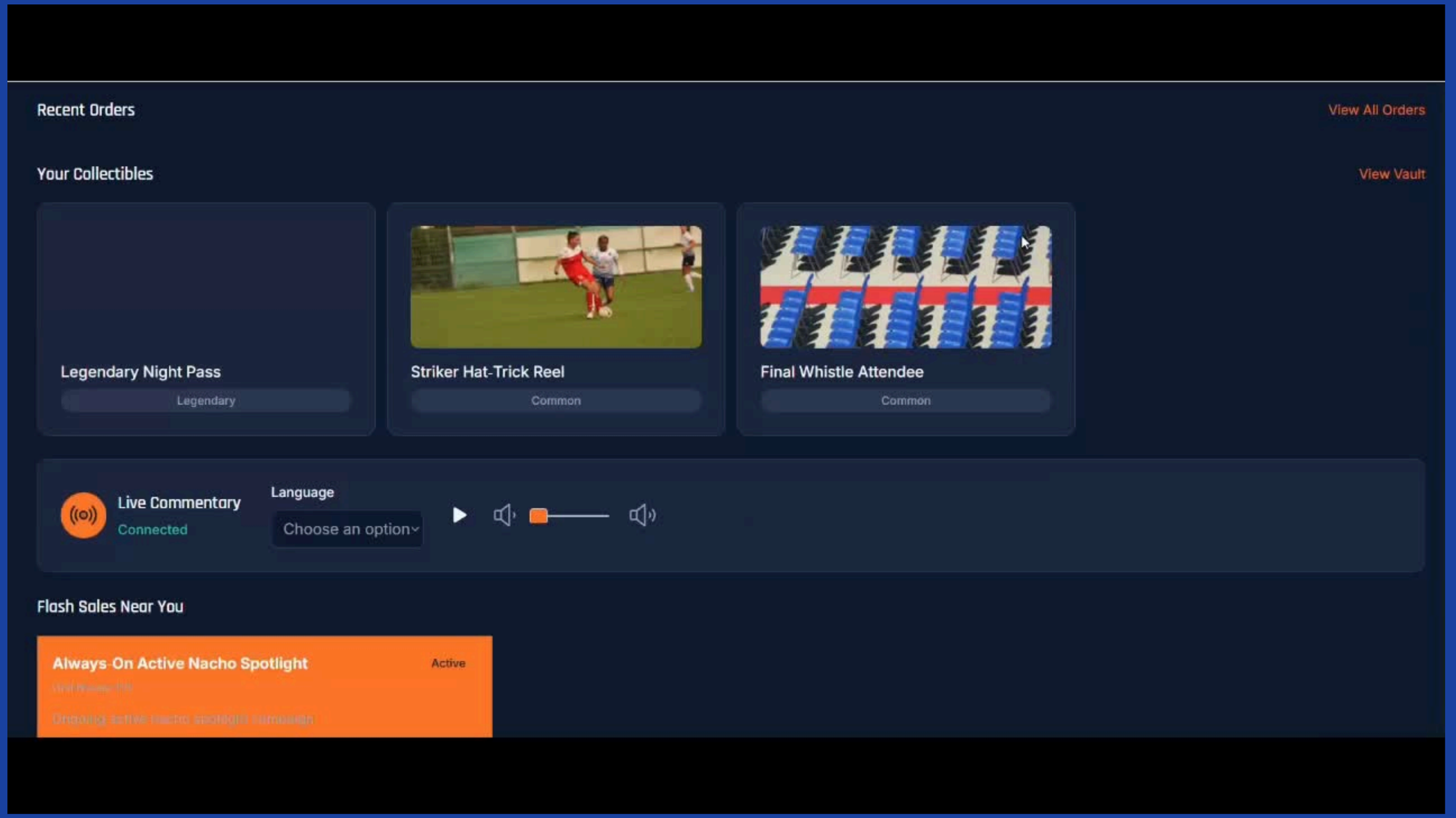
Realism

Phased Implementation Strategy

- Start with the “Sense” layer to collect real-time crowd data
- Gradually roll out “Act” features such as AR navigation
- Allow the system to learn each stadium’s unique crowd patterns
- Continuously refine performance through data-driven adaptation
- Reduce risk through staged, scalable deployment



Prototype


Recent Orders
View All Orders
Your Collectibles
View Vault
Legendary Night Pass
Legendary
Striker Hat-Trick Reel
Common
Final Whistle Attendee
Common
Live Commentary
Connected
Language
Choose an option
Flash Sales Near You
Always-On Active Nacho Spotlight
Active

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Thank you!

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ARENA OS



